

ELECTRIC SHOCK: is the passage of electric current through the human body whenever he/she comes in contact with the source of electricity.

RESPONDING TO ELECTRIC SHOCK:

- Turn the power off
- Turn off the switch
- Pull out the plug
- Remove the live wire with a dry stick, wood or insulated wire cutters

TREATMENT OF ELECTRIC SHOCK

- Check the person's breath, pulse, skin colour and ability to respond to you:
- Immediately use cardio pulmonary resuscitation (CPR) if the person's heart beat has stopped or seems very rapid or irregular.
- Have someone call for emergency medical help.
- Cover all outlets
- All wires should be properly insulated
- Provide adult supervision whenever people are in an area with

Potential electrical hazards.

ARTIFICIAL RESPIRATION

This is any method that causes air to flow in and out of a person's lungs when natural breathing ceases, as in respiratory paralysis, drowning, electric shock, choking gas or smoke inhalation or poisoning.

METHODS OF ARTIFICIAL RESPIRATION

- Mouth - to- mouth
- Mouth- to- nose

RADIATION EXPOSURE

RADIATION: Refers to high speed subatomic particles, such as streams of alpha or beta particles or electromagnetic radiation.

RADIOACTIVITY: Is the spontaneous transformation of an unstable atom and often results into the emission of radiation. This process is referred to as a transformation, a decay or disintegration of an atom.

TEMPERATURE MEASURING DEVICES
Thermal Equilibrium

Compare, contrast
Compare answers

COMMON TYPES OF RADIATION

- Alpha radiation (α)
- Beta (β)
- Gamma (γ)
- Neutrons (n)
- X-rays

NUCLEAR FUSION: is a process where two or more nuclei combine to form an element with a higher atomic number with the release of high amount of energy.



NUCLEAR FISSION: is a process where a larger nucleus splits into two approximately equal nuclei with the release of high amount of energy.



N.B: Ba= barium, Kr= krypton, U= uranium

Example 1

Identify the type of radiation involved in each of the following equations:

- (i) ${}^{234}_{90}\text{Th} \longrightarrow {}^{234}_{91}\text{Pa} + {}^0_{-1}\text{e}$ Beta particle
- (ii) ${}^{234}_{91}\text{Pa}^* \longrightarrow {}^{234}_{91}\text{Pa} + \gamma$ Gamma particle
- (iii) ${}^{238}_{92}\text{U} \longrightarrow {}^{234}_{90}\text{Th} + {}^4_2\text{He}$ Alpha particle
- (iv) ${}^{226}_{88}\text{Ra} \longrightarrow {}^{222}_{86}\text{Rn} + {}^4_2\text{He}$ Alpha

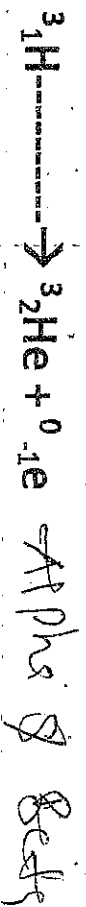
Soln

- (i) Beta radiation
- (ii) Gamma radiation
- (iii) Alpha radiation
- (iv) Alpha radiation

Example 2

If a nucleus of tritium decays and a nucleus of Helium 3 is formed, state the radioactive particle that will be emitted in the reaction process.

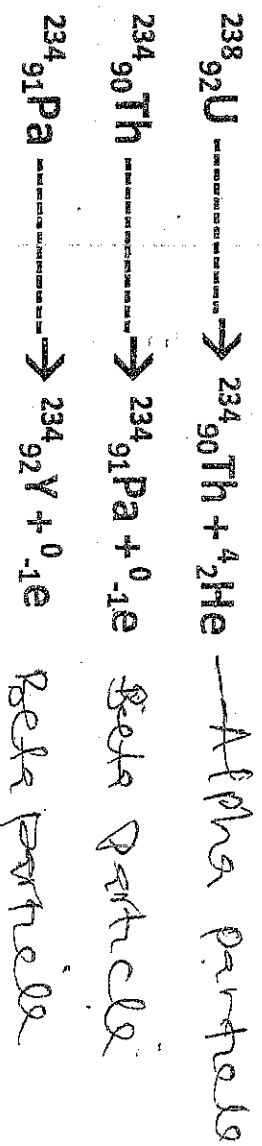
Soln



Example 3

${}^{238}_{92}\text{U}$ is a long half-life alpha emitter and decays to Thorium Th which in turn decays by beta emission with a small decay constant to an isotope of Protactinium Pa. The protactinium isotope finally decays by beta emission to an element Y. Write down the decay scheme of ${}^{238}_{92}\text{U}$ as stated above.

Soln



Balances

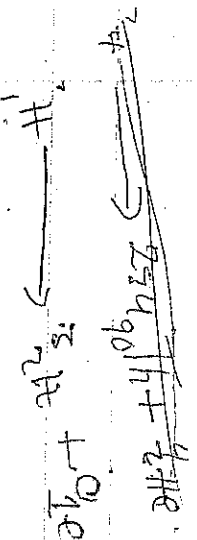
These are instruments used in measuring the mass or weight of bodies.

The balance Room

This is a reserved location in the laboratory where balances are kept and weighing carried out.

TYPES OF BALANCES

- (i) Beam balance
- (ii) Spring balance
- (iii) Substitution balance
- (iv) Analytical balance
- (v) Electromagnetic balance
- Prescription balance



SENSITIVITY OF A BALANCE

This is the ability of a balance to detect very small amount of weight or mass on its pan.

Example

In a Wheatstone bridge, a change of 5Ω in the unknown arm of the bridge is required to produce a change in deflection of 2.5mm of the galvanometer. Find (i) the sensitivity and (ii) the deflection factor.

(i) Sensitivity = magnitude of output response/ magnitude of input.

$$2.5 / 5 = 0.5 \text{ mm}/\Omega$$

(ii) Deflection factor = $1/\text{sensitivity}$

$$1/0.5 = 2 \Omega/\text{mm}$$

WEIGHING

It is generally agreed that pharmaceutical product should be prepared with a low percentage of error. The official compendia allow a tolerance of + or - 5% for most formulae, although greater accuracy may be required for very potent drugs with greater toxicity potential. This same degree of accuracy is expected in all extemporaneously compounded products.

$$\text{Percentage error} = \frac{\text{error of measurement}}{\text{Quantity desired}} \times 100\%$$

Quantity desired

$$\text{Percentage error} = \frac{\text{sensitivity}}{\text{Quantity desired}} \times 100\%$$

Quantity desired

Example 1

A class III prescription balance has a sensitivity of 6mg. what percentage of error would result in weighing 50mg of drug on the balance?

Soln

$$\% \text{ error} = \frac{6\text{mg}}{50\text{mg}} \times (100\%) = 12\%$$

LEAST WEIGHABLE QUANTITY (L.W.Q)

Similarly, we can calculate the smallest quantity that can be weighed on a balance of known sensitivity to maintain a desired level of accuracy. This weight is referred to as L.W.Q

$$L.W.Q = (\text{sensitivity} / \text{percentage error tolerated}) \times 100\%$$

Example

What is the L.W.Q that will result in an error of 5% or less on a class III prescription balance with sensitivity of 6mg?

Soln

$$L.W.Q = (6\text{mg}/5\%) \times 100\% = 120\text{mg}$$

CARE AND MAINTENANCE OF A BALANCE

- (i) When a new balance is purchased, it should be installed by a specialist.
- (ii) Load should be removed when not in use.
- (iii) Knife edges and pans must always be free from dust.
- (iv) Parts should never be lubricated to avoid accumulation of dust.
- (v) Operation manual must always be consulted whenever problems arise.
- (vi) Balances must be serviced from time to time.
- (vii) All sources of vibration must be avoided where a balance is kept

EFFECTS OF HEAT ON BALANCES

Essential parts of balances are made of metals, so they conduct heat easily. Whenever heated, they tend to expand and so the parts are overstretched. It is then necessary to keep the balances in a balance room away from rays of sunlight. There should be no direct weighing of hot objects on the balances.

EFFECTS OF VIBRATION ON BALANCES

Balances are highly sensitive to vibration because it causes defects on the knife edges and makes them crack. This will lower the sensitivity of the balance and then renders it useless. Therefore, a balance must not be installed very close to a

workshop or other noisy areas. There should not be dragging of furniture wherever a balance is kept.

TEMPERATURE MEASURING DEVICES

Temperature is the physical property which determines whether a body will be in thermal equilibrium with another body.

THERMAL EQUILIBRIUM

Two bodies A (hotter) and B (colder) are said to be in a thermal equilibrium when they are brought close to one another, A loses heat and B gains the heat lost by body A, at a point, the bodies A and B attain an equilibrium point where the temperature of A equals that of B. This equilibrium point is called thermal equilibrium of the two bodies.

The apparatus used in the laboratory for measuring temperature is called thermometer.

TYPES OF THERMOMETER

- (i) Liquid -- in -- glass thermometer
- (ii) Constant volume gas thermometer
- (iii) Platinum resistance thermometer
- (iv) Thermistor
- (v) Resistance thermometer
- (vi) Optical pyrometer

Example 1

A thermometer with an arbitrary scale S registers -30°S at the ice point and 90°S at the steam point. Calculate the Celsius temperature corresponding to 60°S .

Soln

$$(\Theta - 0) / (100 - 0) = 60 - (-30) / (90 - (-30))$$

$$\Theta / 100 = 90 / 120$$

$$\Theta = 75^{\circ}\text{C}$$

